STATEMENT OF WORK
EISENHOWER LOCK SHOTCRETE CONSTRUCTION REPAIR

SECTION 1 – Scope

1.1 – Purpose/Introduction/Overview

Each winter the Great Lakes St. Lawrence Seaway Development Corporation (GLS) performs inspections and winter maintenance on Eisenhower Lock. Inspections during previous winter maintenance periods have indicated the need for concrete rehabilitation at Eisenhower Lock. The intent of this project is to perform concrete rehabilitation in these damaged areas, including the demolition of concrete and installation of steel reinforcement as necessary. The Eisenhower Lock is located at 190A and 76 Barnhart Island Road (North and South sides, respectively), Massena, NY 13662.

SECTION 2 – APPLICABLE DOCUMENTS

The documents listed in this section, combined with the other sections of this Statement of Work (SOW), form the contract requirements for the project. These documents are being furnished to the Contractor in electronic format only.

Drawings – Eisenhower Lock:

- 326-C-20/1 - Lock General Plan Elevations & Sections
- 326-C-20/10 - St Lawrence Seaway Robinson Bay Lock North Wall Sta. 704B to 988B Plan and Elevation
- 326-C-20/11 - St Lawrence Seaway Robinson Bay Lock South Wall Sta. 165A to 128B Plan and Elevation
- 326-C-20/12 - St Lawrence Seaway Robinson Bay Lock South Wall Sta. 128B to 429B Plan and Elevation
- 326-C-20/13 - St Lawrence Seaway Robinson Bay Lock South Wall Sta. 429B to 704B Plan and Elevation
- 326-C-20/130 - St Lawrence Seaway Robinson Bay Lock Frames & Covers Location Plan
- 326-C-20/14 - St Lawrence Seaway Robinson Bay Lock South Wall Sta. 704B to 988B Plan and Elevation
- 326-C-20/2 - St Lawrence Seaway Robinson Bay Lock Guide and Guard Walls Monolith Layout
- 326-C-20/23 - Filling & Emptying Culvert North Wall
- 326-C-20/24 - Filling & Emptying Culvert South Wall
- 326-C-20/25 - Filling & Emptying Culvert Details
- 326-C-20/34 - Tainter Valve Recesses Details
- SLS-326-206 - Eisenhower Lock Cutaway View
- SLS-326-300 - Eisenhower Lock Site Map Indicating Concrete Work Area
- 326-C-20/7 - St Lawrence Seaway Robinson Bay Lock North Wall Sta. 165A to 128B Plan and Elevation
- 326-C-20/8 - St Lawrence Seaway Robinson Bay Lock North Wall Sta. 128B to 396B Plan and Elevation
- 326-C-20/9 - St Lawrence Seaway Robinson Bay Lock North Wall Sta. 396B to 704B Plan and Elevation

Reference Photos – Eisenhower Lock:

- Reference Photos.pdf (8 jpgs)

SECTION 3 – REQUIREMENTS

3.1 – Scope Requirements

This project is for the rehabilitation of concrete at Eisenhower Lock in various locations, which includes the demolition of concrete and installation of steel reinforcement as necessary. Photos and reference drawings have been included for a general guide to the existing structures. Areas where concrete rehabilitation is required include, but are not limited to the following:

- CLINs 0002 and 0003: Work related to these CLINs is in the northside filling and emptying culvert, located approximately between Sta. 105.00A and Sta. 30.00B on print 326-C-20/23. The repair area is in the ceiling of the culvert between the upstream most bulkhead recess and the teardrop ramp. This repair continues from previous years repair in the ceiling. The total area of anticipated concrete repair is 70'x12'x12" in the ceiling.
- CLINs 0004 and 0005: Work related to these CLINs is in the southside filling and emptying culvert on print 326-C-20/24. The repair areas are within the ceiling and walls of the culvert tunnel. Anticipated repair areas in the wall include the south wall at the teardrop ramp, approximately between Sta. 105.00A and Sta. 24.00A with an approximate area of 65'x14'x12". The second repair area in the wall is located approximately between Sta. 87.00B and Sta. 128.00B with an approximate area of 15'x15'x12". Anticipated repair areas in the ceiling include 44'x15'x12" located approximately between Sta. 429.00B and 473.00B, near filling and emptying port 12 when counting upstream to downstream.
- CLINs 0006 and 0007: If, while identifying the extents of damaged concrete related to CLINs 0002-0005, it is determined that additional volume(s) of concrete require repair; the Contractor must not proceed without approval from the COR. Work related to these CLINs is for additional volume(s) of concrete to be demolished and installed within Eisenhower Lock's northside and southside emptying and filling culverts after approval to proceed from the COR.

The actual size and depth of repair areas are approximate from previous inspections, and the actual dimensions may vary at the time of execution of work. The Contractor must coordinate with the COR to identify the extents of the actual repair areas prior to the start of demolition. The Contractor must provide all labor, equipment, materials, and costs required to complete this statement of work. All work must be performed in accordance with American Concrete Institute (ACI) Specification for Shotcrete 506.2 and to generally accepted industry standards.

Concrete rehabilitation work will occur in locations throughout Eisenhower Lock. Patches will vary in depth of new concrete throughout. Reference photos attached to this SOW indicate areas of known deteriorated concrete in the repair areas. Additional areas not pictured in these photos may require rehabilitation work. GLS will identify areas with the Contractor after the Contractor has provided means of access to the repair areas. GLS and the Contractor must agree on the pre-demolition and pre-placement volumes prior to demolition and placement respectively to determine the volumes for payment. GLS estimated volume of repairs for each area reflects the volume of shotcrete to be placed. **The Contractor will only be paid for actual volumes of completed concrete demolition and installation at each location. Note that volumes of concrete waste, such as rebound or over spray, will not be paid for.**

The Contractor must complete the required preparation work for concrete rehabilitation in each location. The Contractor will be authorized to commence work in the lock after initial dewatering which is anticipated to be on or about January 12, 2027. **Access to the areas around the upstream miter gates and the vertical lift gate will be restricted until on or about January 19, 2027. Refer to reference drawing 326-C-20-1 for the approximate limits of the restricted access area. Once GLS work in this area**

is complete, the area will be turned over to the Contractor. Under no circumstances will the Contractor be authorized to access or have materials, tools or equipment within the dewatered lock work areas after March 8, 2027. The Contractor must demobilize all material, tools, equipment and debris offsite by no later than March 15, 2027.

The Contractor must provide means for heating work areas as necessary. Partitions and/or vapor barriers must be erected and maintained by the Contractor to isolate the heating areas. The Contractor must be responsible for a safe work environment and provide means to ensure work areas are accessible for work and inspection at all times. The Contractor must submit a project safety plan for approval prior to mobilization for work.

The Contractor's personnel must not drive vehicles or equipment over any recess covers or gratings located on the lock walls, nor set up cranes or outriggers on any recess covers or gratings because they are not designed for these loads. The locations of recess covers and gratings are indicated on the contract reference drawings; however, the Contractor is responsible for verifying the recess cover and grating locations in the field prior to submitting the work site layout plan for approval and commencing work.

The Contractor must take precautions to not damage any GLS equipment, machinery and facilities. This also includes any terrain and vegetation. Any damage to GLS property must be repaired expeditiously and at the expense of the Contractor.

3.2 – Requirements for Shotcrete Placement

This SOW is for the application of concrete, also referred to as shotcrete, which is pneumatically projected at high velocity in accordance with the "dry" placement method where all ingredients are pneumatically conveyed dry to the nozzle for final mixing with water before projection onto the work surface. The shotcrete dry mix must be supplied as a prepackaged single component dry product ready for field application. Mixing and application must be performed in accordance with American Concrete Institute (ACI) Specification for Shotcrete 506.2. Portions of this specification with other special requirements are presented in this section.

3.2.1 **Material Information Submittals:** Submit the manufacturer's product data and installation instructions for each material and ingredient used. Submit the mix design with proportions of all ingredients. Include the manufacturer's Safety Data Sheets as appropriate. The manufacturer of single-component dry mix product must be a company with at least 15-years of experience in the manufacturing and marketing of pre-packaged cementitious repair materials.

3.2.2 **Quality Assurance:** The Contractor must be experienced and qualified to perform the work and each nozzlemen must be certified in accordance with requirements of ACI CP-60 Craftsman Work Book and the ACI 660 Shotcrete Nozzlemen Certification Committee. The Contractor must ensure two (2) qualified nozzlemen are used to perform shotcrete application, due to the limited schedule available to complete work. A test panel of at least 2-feet square by 8-inches deep must be prepared by the Contractor at the site for each day of shotcrete placement (at the Contractor's expense). The test panels must be cored by an approved laboratory (at the Contractor's expense) for visual examination for bond and consolidation and compressive strength determination. Documentation of laboratory testing must be electronically submitted to the Contracting Officer's Representative (COR) within three days of receipt by the Contractor. Pursuant to the test panel findings, the completed work may also be cored by GLS for evaluation of the in-place bond, consolidation and compressive strength.

3.2.3 **Delivery, Storage and Handling:** Deliver prepackaged shotcrete materials and cements in original packaging, labeled with product identification, manufacturer, batch number and shelf life. Store all ingredients in a dry area. Keep ingredients dry and protected from the weather, direct sunlight and freezing. Handle packaged materials in accordance with manufacturer's printed recommendations.

3.2.4 **Materials:** Shotcrete must be KING Shotcrete RS-D1 or approved equal. The shotcrete material must be a microsilica-fortified, air-entrained cement-based mixture of Portland cement, microsilica, air entrainment, pozzolans and graded aggregates. Coarse aggregates must be no larger than 3/8-inch (0.375") in size. The shotcrete must comply with ASTM C 1480 Type FA. Grades GU, SR, LP. Water- cement ratio of the completed mix must be no greater than 0.40. The 28-day compressive strength must be at least 5,000 pounds per square inch (psi) as tested by an approved laboratory. The Contractor must be responsible for obtaining testing and for associated costs to do so. Air content must be 6 to 9 %. To reduce dust, pre-dampen pre-packaged shotcrete materials before they are conveyed to the nozzle. Instructions provided by the manufacturer regarding mixing and handling must be followed. Use only potable water in shotcrete.

3.2.5 **Surface Preparation:** The repair areas must be dewatered as necessary to allow unobstructed dry access to all surfaces. Additionally, repair areas must be dust/debris free and dampened prior to shotcrete placement. Weeps and/or 1-inch PVC pipes must be placed prior to application to drain water in repair areas as directed by the Contracting Officer Representative (COR). The Contractor must perform all preparation to include squaring off repair areas, saw-cutting to sound concrete and installation of reinforcing mesh and reinforcing bars. Do not use bonding agents. Saw cut the perimeter of each repair area to a depth of at least 1-inch to provide a clean and un-feathered edge. Using low energy hand or light mechanical methods, chip out and remove all unsound concrete. Removal must be carried to sound concrete. The typical removal depth will vary, but could go as deep as 18-inches. Removal depths and locations must be as directed by the COR. All demolished concrete must be removed from the work location and disposed of in the Eisenhower Lock spoil area.

3.2.6 **Reinforcement:** After demolition is completed, for removal depths less than 6"- (relative to the finished surface), install welded wire mesh (WWM) 6 x 6 x W2.9 x W2.9 in the repair area and secure this mesh 12" on center with ¼" J-Style Hook Bolts as manufactured by Ameribest or approved equal, installed 12" on center each way. Use mesh pieces that are as large as practical. Lap adjoining mesh pieces at least one full mesh and lace the splices with wire. For removal depths greater than 6" (relative to the finished surface), install grid of reinforcing bar spaced 12" on center. All grid reinforcing bars must be epoxy coated No. 6, Grade 60. Tie the grid to dowels spaced 3' on center with a minimum embedment of 1'-4" into sound concrete. All dowels to be No. 8 Grade 60 epoxy coated deformed steel reinforcement bar. Embed all dowels with a suitable epoxy adhesive (HILTI HIT-HY 150 MAX or equal). A cover distance of at least 2" of concrete over all embedded steel is required.

3.2.7 **Grade and Alignment:** The proper surface grade and alignment of the completed shotcrete must be maintained by use of taut "ground" wires that are set to the original profile of the concrete base. The shotcrete must then be installed to the ground wires. Ground wires must be installed and spaced no more than 24" on-center to ensure a square and even conformity to the original dimensions of the concrete base. Ground wires must also be set at corners to ensure that the reconstructed corner conforms to the original geometry and dimensions.

3.2.8 **Application:** Place shotcrete first into corners, recesses and other areas where rebound or overspray cannot escape easily. Hold the nozzle perpendicular to the receiving surface and move it in circles to improve consolidation and reduce rebound (separated aggregate that bounces or ricochets out of the placement). In corners, direct the nozzle at a 45-degree angle, or bisect the corner angle if different than 90 degrees. Rebound must not be incorporated in the completed work. Apply the shotcrete in such a manner as to prevent sags and sloughing. Apply around reinforcing materials to completely encase it and have a free flow of material completely around the reinforcement so there are no voids. The front face of reinforcement must remain clean during the encasement. Shotcrete must not be deposited into standing or running water.

3.2.9 **Weather Conditions:** Shotcrete application must not be done in windy conditions that cause separation of the mix. The prevailing ambient temperature must be maintained at least 50 degrees Fahrenheit. Surfaces against which shotcrete is applied must not be below 40 degrees Fahrenheit and must be free of frost. In the event the lower ambient and/or surface temperatures/conditions cannot be achieved, the Contractor must provide means for suitable heat application. Shotcrete materials must be no less than 50 degrees Fahrenheit or more than 100 degrees Fahrenheit, and surfaces against which shotcrete is applied must be no greater than 100 degrees Fahrenheit.

3.2.10 **Finishing and Curing:** Finish immediately after shotcrete application with a steel trowel to a smooth surface where necessary, with care to prevent loosening of the bond. Shotcrete application areas must be cured at least three (3) days in accordance with Cast-in-Place concrete industry standards and, at a minimum, be maintained at an ambient temperature of 50 degrees Fahrenheit or greater. A gun finish is acceptable. However, GLS will utilize a straight edge to ensure no peaks/valleys and/or rolling surfaces. The straight edge (approximately 8’ in length) will be placed against the finished product in random positions and there must be no more than 1/2” variance across the finished product.

3.2.11 **Disposal:** Contractor must dispose of all materials, debris, rebound, packaging and any other items used in performance of this SOW. All items must be hauled offsite for disposal and/or recycling. However, rebound demolished concrete and excess aggregate can be disposed of at Eisenhower Lock as directed by the COR. All waste must be quantified by weight and submitted to the project COR prior to final payment.

3.3 – Task Requirements

Task 1 – Review construction/installation with GLS, provide project deliverables. Mobilize all equipment and materials to site.

Task 2 – Perform demolition of deteriorated concrete and place steel reinforcement in areas of the Eisenhower Lock northside filling and emptying culvert per the SOW.

Task 3 – Install new concrete in areas of the Eisenhower northside filling and emptying culvert per the SOW.

Task 4 – Perform demolition of deteriorated concrete and place steel reinforcement in areas of the Eisenhower Lock southside filling and emptying culvert per the SOW.

Task 5 – Install new concrete in areas of the Eisenhower Lock southside filling and emptying culvert per the SOW.

Task 6 – Demobilize all equipment and materials from the site.

3.4 – Deliverables

Deliverable	Due Date/Frequency	Submit To:
Attend pre-construction meeting	**Within 5 days of Award**	**N/A**
Construction Schedule	**Within 5 days of Award**	**COR**
Schedule of Values	**Within 5 days of Award**	**COR**
Submittal Log in accordance with SOW	**Within 5 days of Award, prior to ordering materials**	**COR**
Project Safety Plan	**Within 5 days of Award**	**COR**
Worksite Layout Plan	**Within 5 days of Award**	**COR**
Certified Payroll	**Weekly**	**CO**
Coordinate and host weekly progress meetings and provide meeting minutes	**Weekly**	**COR**

3.5 – General Conditions

3.5.1 – Period of Performance

Period of Performance is as follows:

Anticipated Schedule	
Lock Dewatered	January 12, 2027 (on or about)
Complete Work Within Lock	March 8, 2027
Complete Work Outside Lock and Demobilize	March 15, 2027

3.5.2 – Place of Performance

All work must take place at Eisenhower Lock, located at 76 and 190A Barnhart Island Road, Massena, New York, 13662.

3.5.3 – Government Furnished Equipment/Information/Utilities

Process water, potable water, heat, air, and any other utilities needed for this SOW must be the responsibility of the Contractor. All temporary facilities, including restroom facilities, must be provided by the Contractor. The GLS will provide the Contractor with two (2) load centers which will be located on top of the lock wall or other locations agreed upon by the Contractor and COR. There are physical limitations of where the load centers can be placed and still be able to be energized. Historically, there has been unscheduled temporary power loss to GLS facilities. Therefore, the Contractor must make provisions and plan accordingly. Each load center will provide the Contractor with a total electrical capacity of 60 amps of three-phase 208/120 volt power. The load centers each have six (6) 120V single phase twist-lock receptacles, 20 amp each, two (2) standard 120V single phase receptacles, 20 amp each, two (2) 208V three-phase twist-lock, 30 amps combined, and one 208V single phase twist-lock at 20 amps. All conductors, lights and outlets from there must be provided by the Contractor. The Contractor must provide any additional power required at his/her own expense. Temporary wiring, extension cords

and GFCI protection devices must be properly sized and conform to applicable provisions of OSHA and the National Electrical Code. GLS will provide a maximum of two electric load centers concurrently. Any additional electrical power must be provided by the Contractor at his/her own expense. Additionally, the GLS will not provide housing/break rooms and the contractor will be responsible for any and all utilities/facilities aside from the load centers indicated above. The Contractor must provide all remaining construction materials to complete the project.

3.5.4 – Facilities Access

All locations are normally submerged in water. Demolition and construction work can only be done when the lock is out of service and dewatered during the annual winter navigation shutdown period (season). Access to the work areas is restricted in that the filling and emptying culverts are below the adjacent lock wall surfaces. This will require temporary special access means (at the Contractor's expense) such as construction stairways, elevators, etc. for personnel access, and crane service (at the Contractor's expense) will be required to insert and remove materials and equipment. Equipment access to the filling and emptying culverts can be made through the upstream culvert (tainter) valve recesses. All equipment lowered into the culvert valve recesses will need to fit through the culvert valves. This opening is approximately 15'x8'-6" and requires advanced coordination with GLS to maneuver the culvert valve. GLS cannot fully dewater the work areas. Additionally, water will infiltrate into the dewatered areas. Water levels are anticipated to fluctuate by as much as approximately 4-feet in the work areas. Therefore, the Contractor must provide means to manage water in the work area as needed.

3.5.5 – Responsibilities

GLS will be responsible for the following:

- Dewatering the lock chamber (as described in this SOW)
- Providing up to two electrical load centers (as described in this SOW)

The Contractor must be responsible for the following:

- Personnel, materials, and equipment to complete the work.
- Any utilities required to complete the work. Utilities include, but are not necessarily limited to, water, heat, electricity, restroom facility and lighting.
- Contractor must provide any crane and support equipment/labor.

3.5.6 – Environment Protection

3.5.6.1 **General:** For the purpose of this specification, environment protection is defined as the retention of the environment in its natural state to the greatest possible extent during project construction and the enhancement of the natural appearance in its final condition. Environment protection requires consideration of air, water and land resources and involves noise, solid waste management and management of liquid and gaseous pollutants, radiant energy and radioactive materials as well as other pollutants. In order to prevent and provide for abatement and control of any environmental pollution arising from the construction activities in the performance of this contract, the Contractor and his/her subcontractors must investigate and comply with all Federal, State and local laws and regulations concerning environmental pollution control and abatement. Any non-compliance must be immediately corrected and the GLS notified.

3.5.6.2 **Notification:** The COR will notify the Contractor whenever the GLS detects any noncompliance with the aforementioned Federal, State and/or local laws and regulations. Such notice, when delivered to the Contractor or his/her authorized representative at the site

of the work, must be deemed sufficient for the purpose. The Contractor must, after receipt of such notice, immediately inform the COR of the proposed corrective action and take such action as may be approved. If the Contractor fails or refuses to comply promptly, the COR may issue an order stopping all or a part of the work until satisfactory corrective action has been taken. No part of the time lost due to any such stop order will be made the subject of a claim for extension of time or for excess costs or damages incurred by the Contractor.

3.5.6.3 **Protection of Land Resources:** The land resources within the project boundaries and outside the limits of permanent work performed under this contract must be preserved in their present condition or be restored to a condition after completion of construction that will appear to be natural and not detract from the appearance of the project. The Contractor must confine his/her activities to areas defined by the contract plans and specifications.

3.5.6.4 **Protection of Water Resources:** The Contractor must not pollute the ground or surface waters with fuels, oils, bitumen, calcium chloride, acids, insecticides, herbicides or other harmful materials. The Contractor must investigate and comply with all applicable Federal, State and local laws and regulations concerning pollution. The Contractor must complete a thorough cleanup of work areas prior to March 8, 2027 to prevent debris polluting the river when GLS re-floods the lock.

3.5.6.5 **Permits and Insurance:** The Contractor must be responsible for researching and obtaining all permits that may be required for the work. The Contractor must also have insurance as appropriate for the class of work.

3.5.7 – Plant

The Contractor must furnish all plant required for the performance of the contract. The plant must be in satisfactory operating condition and capable of safely and efficiently performing the work. The plant will be subject to the inspection of the COR at all times.

3.5.8 – Use of Explosives

Blasting will not be permitted.

3.5.9 – Safety

The Williams-Steiger Occupational Safety and Health Act of 1970 (as amended) as set forth in 29 CFR 1926/1910, will govern this contract. The Contractor must be responsible for a safe work environment at all times. The Contractor must comply with requirements of OSHA. The Contractor must submit a safety plan for approval prior to commencing work.

3.5.10 – Other Work

Maintenance work by GLS personnel and other work by other contractors may be in progress concurrently with the work under this contract. The Contractor must take measures to prevent an unsafe or unhealthful environment for other workers. Only the Contractor's authorized personnel, subcontractors and suppliers must be allowed to enter the work area. The Contractor must fully cooperate with the GLS in all phases of the work and any disputes which may arise will immediately be brought to the attention of the Contracting Officer through the COR.

3.5.11 – Security and Background Investigations

The Contractor shall commence coordination with the GLS Chief of Security immediately upon contract award to initiate the background investigation and badging process, as these reviews require significant

processing time. All regular on-site contractor and subcontractor employees must obtain security clearance prior to being granted unescorted physical access to GLS facilities.

As part of the background investigation submission, the Contractor must provide a current resume for each employee requiring access. The Contractor must strictly comply with Transportation Acquisition Regulation (TAR) Clauses 1252.204-70 (Contractor Personnel Security and Agency Access) and 1252.237-70 (Qualifications of Contractor Employees) as stated within the RFQ.

GLS may, at its sole discretion, provide escorts for contractor employees whose clearances are pending; however, it remains the Contractor's ultimate responsibility to ensure timely coordination and submission of all required security documentation to avoid schedule impacts. Fingerprinting can be completed locally in Massena, NY, at no cost to the Contractor. Alternatively, GLS can provide fingerprint cards for completion at a facility of the Contractor's choice, with any associated fingerprinting costs borne by the Contractor.

3.5.12 - Weekly Progress Meetings
The Contractor shall coordinate and hold weekly progress meetings to include attendance by the Contractor, Contracting Officer, and COR. The Contractor shall provide updates to work progress, planned work, revisions to the construction schedule with estimated completions and actual completions. The Contractor shall record meeting minutes and distribute draft minutes within two calendar days of each meeting.

3.5.13 – Contracting Officer's Representative
The Contracting Officer's Representative (COR) monitors all technical aspects of the contract and assists in contract administration. The COR ensures that the Contractor performs the technical requirements of the contract, performs inspections necessary in connection with contract performance, maintains written and oral communications with the Contractor concerning technical aspects of the contract, and monitors Contractor performance.